GABRIEL ASAEL TARIGAN

Curriculum Vitae

Lecturer Specialist • Artificial Intelligence • Machine Learning

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PROFILE

Applied machine learning lecturer and researcher in the Artificial Intelligence Program at BINUS University's School of Computer Science. I design and teach the program's AI courses, and I build and operate the machine-learning evaluation and assessment platforms that students and lecturers use every day. My research is applied: computer vision for observable student engagement, time-series forecasting, and applied LLM and NLP studies, with 11 published papers and 8 further papers accepted at 2026 conferences. Master of Computer Science, Summa Cum Laude (GPA 3.95/4.00).

Research interests: applied machine learning; computer vision and pose estimation for education; time-series forecasting; machine-learning systems and MLOps.

EDUCATION

BINA NUSANTARA UNIVERSITY — **Jakarta**
Master of Computer Science (M.Kom.). **GPA 3.95/4.00, Summa Cum Laude** — 2023–2024
Thesis: Lighter student engagement recognition in a classroom environment using skeletal keypoints. Supervisor: Prof. Dr. Bens Pardamean. Faculty Development Program (FDP) Scholarship.

SOEGIJAPRANATA CATHOLIC UNIVERSITY — **Semarang**
Bachelor of Computer Science. **GPA 3.79/4.00, Best Graduate** — 2018–2022
Thesis: Illness prediction from oral symptoms using machine learning with a small dataset.

EXPERIENCE

BINA NUSANTARA UNIVERSITY, School of Computer Science, Artificial Intelligence Program — **Jakarta**
Lecturer Specialist — 2025–Present
Subject Developer & Researcher — 2024–Present

- **Curriculum.** Authored 5 and co-authored 2 courses spanning machine learning, deep learning, applied NLP, computer vision, MLOps, signal processing, and IoT and robotics, each with course outline, assessment design and lab materials. Teach at undergraduate and graduate level.
- **Platforms.** Designed, built and operate ARISE, a large-scale machine-learning evaluation and benchmarking platform with live leaderboards and automated evaluation (about 30 submissions per day), alongside automated assessment portals and interactive learning tools. These systems are in daily production use by students and lecturers (Python, FastAPI, PyTorch).
- **Research and supervision.** Conduct applied research in computer vision and time-series forecasting. Supervised 8 undergraduate capstone projects in 2026 (computer vision, hoax detection, sign-language recognition, medical screening, assistive navigation).
- **Competitions and outreach.** Coordinate two national AI competitions (ARISE, AIPLEX) and lead AI-literacy outreach across Indonesia.

PUBLICATIONS

11 published (3 journal articles, 8 conference papers), 8 accepted for 2026 conferences, 1 preprint. First-author papers: J2, J3, C2, C7. Also on Google Scholar and ORCID.

JOURNAL ARTICLES

J1 Gunawan, A. A. S., Irwansyah, E., **Tarigan, G. A.**, Tedjasulaksana, J. J., Juliastuti, & Wijayanti, Y. (2026). Assessing nature-based solutions for Jakarta flood risk reduction through integrated existing and projected flood hazard mapping. *Evergreen*, *13(3)*, *1251–1266*. doi:10.5109/7442747

J2 **Tarigan, G. A.**, Nugroho, K. S., & Pardamean, B. (2026). Skeletal keypoint-based pipeline as a computer vision-based approaches. *Communications in Mathematical Biology and Neuroscience*. doi:10.28919/cmbn/9820

J3 **Tarigan, G. A.**, Elwirehardja, G. N., Nugroho, K. S., & Pardamean, B. (2025). Lighter student engagement recognition in a classroom environment using skeletal keypoints. *IAENG International Journal of Computer Science*, *52(6)*, *1997–2014*.

Conference papers (published)

C1 Harun, J., Yulianto, D., Andrywinata, C., Hermawan, E. S., Pranoto, H., et al. (2025). Product sales recommendation system using item-based collaborative and content-based filtering. *ICCSCI 2025*, *Procedia Computer Science*, *269*, *1643–1653*. doi:10.1016/j.procs.2025.09.107

C2 **Tarigan, G. A.**, Elwirehardja, G. N., Nugroho, K. S., & Pardamean, B. (2025). Computer vision-based pose estimation for student engagement detection: Trends and challenges. *2025 7th International Conference on Cybernetics and Intelligent System (ICORIS)*, *IEEE*, *1–5*. doi:10.1109/icoris67789.2025.11296082

C3 Prastyapradipta, B., Naoko, K., Himawan, R. A. B., Ibrahim, M. A., & **Tarigan, G. A.** (2025). Analyzing Indonesian political hoax detection system on social media using deep learning and natural language processing. *ICCSCI 2025*, *Procedia Computer Science*, *269*, *1742–1751*. doi:10.1016/j.procs.2025.09.117

C4 Boer, Y., Permatasari, A. C., Kevin, K., **Tarigan, G. A.**, & Suhartono, D. (2025). InterQ: A job interview training platform using facial expression detection and answer scoring. *2025 8th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI)*, *IEEE*, *841–846*. doi:10.1109/isriti68345.2025.11393150

C5 Nasari, M., **Tarigan, G. A.**, Masaling, N. A. P., Joddy, S., & Minor, K. A. (2025). Indonesian food detection with YOLO and calorie estimation using lightweight LLMs. *2025 8th ISRITI*, *IEEE*, *784–789*. doi:10.1109/isriti68345.2025.11393164

C6 Perdana, G. A., Wijaya, I. I., Fahreza, K. A., & **Tarigan, G. A.** (2025). Ablation study: Calibrated adaptive learning ensemble methodology. *ICCSCI 2025*, *Procedia Computer Science*, *269*, *993–1001*. doi:10.1016/j.procs.2025.09.041

C7 **Tarigan, G. A.**, Hermawan, E. S., & Girsang, A. S. (2024). Parallelization of LSTM-GRU architectures for multivariate prediction of stock prices. *2024 International Conference on Information Management and Technology (ICIMTech)*, *IEEE*, *311–315*.

C8 Nasari, M., Pradana, R. C., **Tarigan, G. A.**, Masaling, N. A. P., & Tedjasulaksana, J. J. (2024). Enhancing node classification: Integrating particle swarm optimization with graph convolutional networks. *2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)*, *IEEE*, *1239–1243*. doi:10.1109/icicyta64807.2024.10912873

Accepted for presentation (2026)

A1 Taslim, S. V., Amethyst, M. G., **Tarigan, G. A.**, Nugroho, K. S., & Pardamean, B. (2026). Emotional framing effects on medical triage in general-purpose and medical-adapted LLMs. *ICICyTA 2026*, *IEEE*.

A2 Yovan, K., **Tarigan, G. A.**, & Sutoyo, R. (2026). Enhancing cross-dataset generalization in deepfake detection: A multi-source fine-tuning approach with XceptionNet. *2026 9th ISRITI*, *IEEE*.

A3 Putra, R. D., Kumala, C. D., Salim, J. S., Sutoyo, R., & **Tarigan, G. A.** (2026). Application of augmented reality for motorcyclist's sitting posture guidance. *ICICyTA 2026*, *IEEE*.

A4 Chandra, V. P., Kwek, W. V., Tjandra, A. A. M., **Tarigan, G. A.**, & Sutoyo, R. (2026). A low latency Python-Unity integrated framework for real-time BISINDO recognition: A comparative analysis of coordinate-based deep learning architectures. *2nd International Conference on Artificial Intelligence for Sustainability in the Developing World (AISDW)*.

A5 Sutanto, A. A., Pariandika, I. G. W. D., Zefanya, K., Muliana, R. A., **Tarigan, G. A.**, & Samuel. (2026). WASPADA (Weapon And Suspicious Pose Alert & Detection Aggregator) to enhance ATM security via CCTV. *1st International Conference on Computing and Digital Technology (ICoDiT)*.

A6 Sunoto, D. N., Febrianto, Frolentika, Davalynda, J., Sulistiawati, W., & **Tarigan, G. A.** (2026). EfficientNet-B0 with batch-hard triplet loss fine-tuning for multi-input signature forgery detection. *ICoDiT 2026*.

A7 Maulina, A., Widhiarto, E., Charisma, R. A., **Tarigan, G. A.**, & Minor, K. A. (2026). Comparative evaluation of CNN, Vision Transformer, and hybrid architectures with Grad-CAM explainability for vision-based malware classification. *ICoDiT 2026*.

A8 Wijaya, D., Deviana, F., Jonowidjojo, G., Yap, S. S., Sihombing, T. S., & **Tarigan, G. A.** (2026). Hybrid deterministic-generative framework: Integrating financial risk classification and narrative-driven portfolio optimization. *ICoDiT 2026*.

Preprint

P1 Laychi, S., Karuntu, S. P. A., Sihombing, D. H., Sutoyo, R., & **Tarigan, G. A.** (2026). Fake review detection on e-commerce platforms using a hybrid anomaly detection approach: Combining autoencoder and isolation forest. *Research Square (preprint)*. doi:10.21203/rs.3.rs-10162747/v1

ACADEMIC SERVICE AND LEADERSHIP

EMACS Conference, Peer Reviewer 2024–Present
Review submitted manuscripts for technical soundness, methodological rigour, originality and relevance, and provide constructive feedback.

ICCSCI (International Conference on Computer Science and Computational Intelligence), Session Chair 2024–2025
Chaired conference sessions and handled opening duties.

ARISE, Coordinator 2026
Coordinate a national machine-learning competition for students, run on the custom evaluation platform with live leaderboards and automated scoring.

AIPLEX (AI Prototype and Logic Exhibition, BEEFEST), Coordinator 2025
Developed and coordinated a national AI competition for high school students end to end, themed around AI for everyday problems and aligned with the SDGs.

COMMUNITY OUTREACH

Volunteer, Indonesia Present

- AI and programming workshops for eight schools (SMA Karangturi, SMAN 82, SMA Immanuel, SMA Saint John, SMA Narada, SMA Kesatuan, SMA Yakobus, Binus School Bekasi) and regional workshops in Pontianak and Palembang.
- Extracurricular programming programs and community education events at BINUS Alsut (ICONATION); AI learning materials for GreatNusa.
- Generative AI training for small-business community development (CommDev UMKM).

HONOURS AND AWARDS

- **Faculty Development Program (FDP) Scholarship** for master's studies, BINUS University.
- **Summa Cum Laude**, Master of Computer Science, BINUS University (2024); **Best Graduate**, Bachelor of Computer Science, Soegijapranata Catholic University (2022).
- **Bangkit Academy 2022**, Machine Learning path (Google, GoTo, Traveloka); led "Kerjamin", the team's capstone project, a community-driven freelancing platform to widen income opportunities for rural workers.

TRAINING AND CERTIFICATIONS

- **SecOps AI-ML Pen Certified Professional** (2025): practical exam on AI/ML security, including prompt injection and RAG poisoning.
- **Microsoft AI-900** (2025): core AI and ML concepts and the Azure AI service stack.
- **Kominfo Digital Talent Scholarship**, AI track by Huawei (2021), with the **Huawei Certified ICT Associate, Artificial Intelligence** credential.

SKILLS

Research and modelling: Python for data processing, model development, experimentation and evaluation; PyTorch, TensorFlow, scikit-learn, NumPy, pandas, OpenCV.
ML systems: MLOps, model deployment, automated evaluation, FastAPI, REST APIs, SQL, Docker, GitHub Actions, Linux.
Other: Java, PHP, C, JavaScript, HTML/CSS; LLM APIs and AI coding assistants for development.

LANGUAGES

Bahasa Indonesia: native. English: EF SET C2 (Proficient).